

CSE251: ASSIGNMENT-2

Full Marks: 70

1. [CO3] **Design** a circuit using OP-Amp that takes V_{in} as the input and generates V_{out} as illustrated in Figure-1

- I. Design the circuit using inverting amplifier .
- II. Design the circuit using non-inverting amplifier .

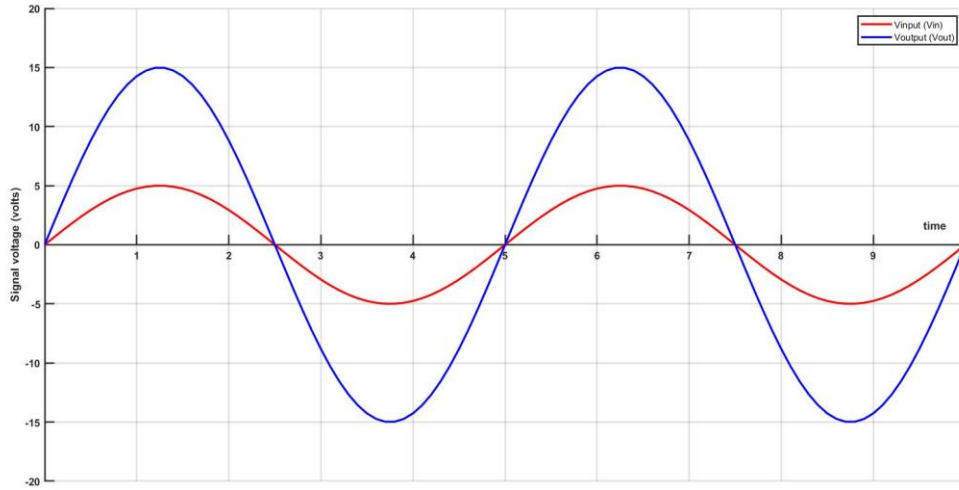


Figure 01

2. [CO-3] **Design** a circuit using OP-Amp to generate the output V_{out} from the input V_{in} as illustrated in figure-2 .

[Hinst: use summing amplifier]

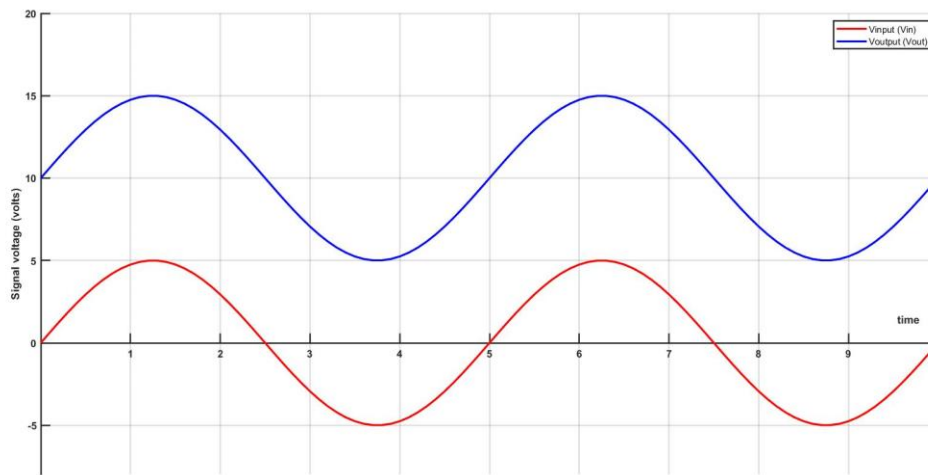


Figure 02

3. [CO3] **Design** a circuit with OP-Amp to implement the following functions

- I. $f = 5x - 10 \frac{dy}{dt} + 2 \int z dt$
- II. $f = \int x dt - 2 \frac{dy}{dt} - z$

4. [CO2]

Consider the circuit shown in Figure 3 Determine the diode current I_D and diode voltage V_D for

- $V_{D0} = 0.6\text{V}$
- $V_{D0} = 0.7\text{V}$

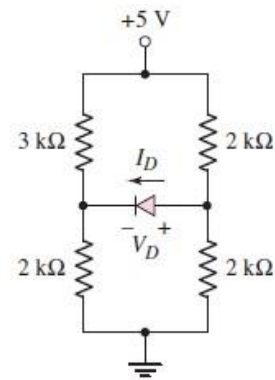


Figure 03

5. [CO2]

Analyze the circuit shown in figure-04 to Find I_{D1} and I_{D2} for

- $R = 100\text{k}\Omega$ and
- $R = 0.02\text{k}\Omega$

Use CVD model with $V_{D0} = 0.7\text{V}$.

[Validate assumptions]

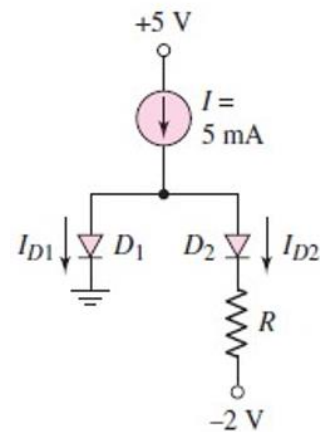


Figure 04

6. [CO2]

- Analyze** Figure-5 to determine boolean expression in terms of V_1 , V_2 , V_3 and V_4 .
- Determine** the values of V_{O1} , V_{O2} and V_O , where $V_1 = 2$, $V_2 = 3$, $V_3 = 2.5$, $V_4 = 3.5$. Assume $V_{D0} = 0.6$ for each diode.

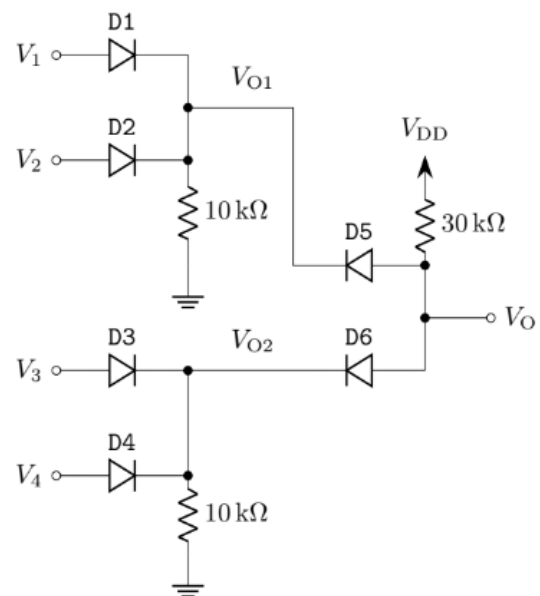


Figure 05

7. [CO1] **Draw** the I-V characteristics of a diode for Ideal, CVD and CVD+R model and show equivalent circuit for each model.